

# 02 Architecture

Decision boundary, data flow and implementation scope

## 1. Core architectural boundary

The primary architectural decision is to keep probabilistic language-model work outside the trusted decision boundary. LangGraph and Llama may interpret natural language and explain a sealed receipt, but the deterministic RAMIFY backend creates product truth, applies policy, authorises actions and signs receipts.

## 2. Data flow

| Stage                  | Responsibility                                  | Authority                                 |
|------------------------|-------------------------------------------------|-------------------------------------------|
| User / Shop            | Product selection or free-form request          | No trust authority                        |
| Interpretation Adapter | Exact match, safe matcher, or LangGraph + Llama | Can suggest known catalogue subject only  |
| RESOLVE                | identify, resolve and status                    | Deterministic identity and standing       |
| RATIFY                 | verify evidence and rules                       | Deterministic evidence findings           |
| Agent Policy           | Narrow the objective result                     | May make stricter, never more permissive  |
| Action Gate            | Choose permitted simulated action               | Bound to signed receipt                   |
| Receipt Ledger         | Append local synthetic records                  | Verifiable history and successor receipts |

### 3. Five primitives

| Primitive | Input                                   | Output                                                              | UI presentation                                   |
|-----------|-----------------------------------------|---------------------------------------------------------------------|---------------------------------------------------|
| identify  | GTIN, SKU, subject_ref or request match | one subject or unresolved                                           | primitive card with input/output and milliseconds |
| resolve   | subject_ref                             | canonical product record                                            | product passport and stage trace                  |
| status    | subject_ref                             | advisory or recall standing                                         | status chip and risk path                         |
| verify    | subject + policy + evidence             | seven detailed check results                                        | evidence trail and check list                     |
| assess    | product truth + actor policy            | objective posture, agent decision, permitted action, signed receipt | objective-to-agent handoff and receipt            |

### 4. Scope note

The signed RESOLVE v0.4 brief described a minimal Go binary proof. This release is the expanded Python/FastAPI browser demonstration. The important carry-over is the deterministic RESOLVE/RATIFY boundary and the independent receipt-verification story; the implementation language is documented here as deliberate prototype evolution rather than hidden drift.